The NEW ENGLAND JOURNAL of MEDICINE

Daily Duration of Oxygen Therapy for Severe Hypoxemia

A PLAIN LANGUAGE SUMMARY

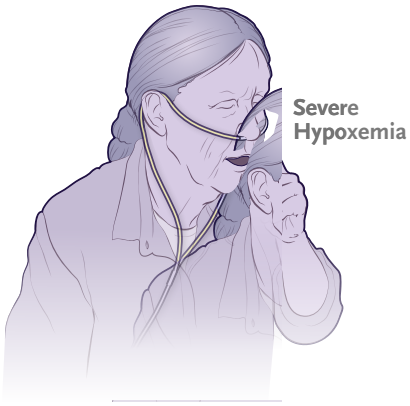
Based on the NEJM publication: Long-Term Oxygen Therapy for 24 or 15 Hours per Day in Severe Hypoxemia by M. Ekström et al. (published September 10, 2024)

In this trial, researchers assessed hospitalization and mortality among patients with chronic severe hypoxemia who received long-term oxygen therapy for either 24 or 15 hours per day.

Chronic severe hypoxemia, or low oxygen levels in the blood, can affect people with chronic obstructive pulmonary disease (COPD) and other respiratory conditions and is associated with poor survival. Long-term oxygen therapy improves survival but is burdensome, and the daily duration of supplemental oxygen needed to improve survival is not known.

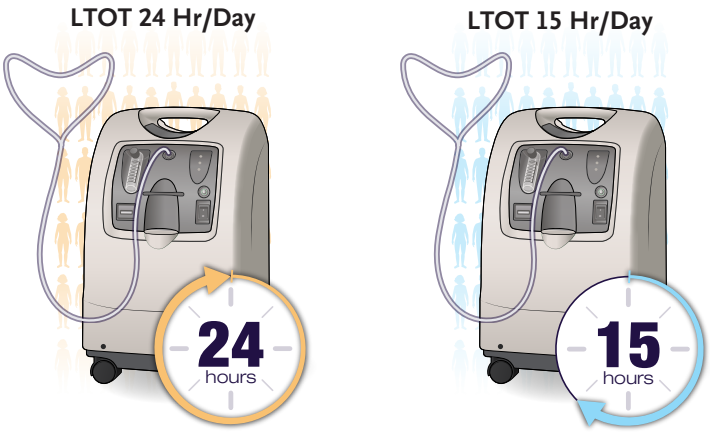
WHY WAS THE TRIAL DONE?

Data from nonrandomized comparisons have suggested that survival might be better with long-term oxygen therapy used for 24 hours per day than with the typically recommended 15 hours per day. Data from randomized trials directly comparing these two durations of oxygen therapy are lacking.



HOW WAS THE TRIAL CONDUCTED?

241 adults who were starting oxygen therapy for chronic severe hypoxemia at rest were randomly assigned to receive long-term oxygen therapy (LTOT) for either 24 or 15 hours per day. The primary outcome was a composite of hospitalization or death from any cause within 1 year.



117 Patients

124 Patients

PATIENTS

WHO

241 adults

Mean age, 76 years

Women: 59%; Men: 41%

CLINICAL STATUS

Chronic severe hypoxemia at rest

Primary diagnoses: COPD in 71%; pulmonary fibrosis in 14%

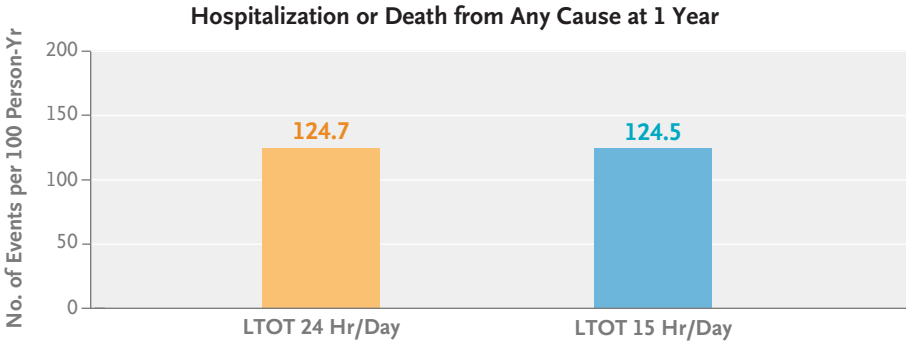
TRIAL DESIGN

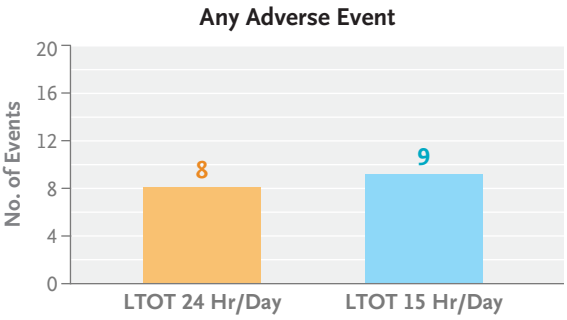
- PHASE 4
- MULTICENTER
- REGISTRY-BASED
- RANDOMIZED
- CONTROLLED
- LOCATION: 20 RESPIRATORY OUTPATIENT CLINICS IN SWEDEN

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RESULTS

At 1 year, the rate of hospitalization or death from any cause did not differ significantly between the two treatment groups.





The number of adverse events, including burn injuries, fall-related injuries, and nosebleeds, was similar in the two groups.

ADHERENCE



At 12 months, patients reported closely adhering to their assigned daily duration of treatment.

LIMITATIONS AND REMAINING QUESTIONS

- Patients reported the daily duration of their oxygen use, so it is possible that adherence to the assigned daily duration of treatment may have been overreported.
- Because of insufficient recruitment, the primary outcome was changed from death to a composite outcome of hospitalization or death from any cause after 110 patients had undergone randomization.
- A benefit of long-term oxygen therapy used for 24 hours per day with respect to the risk of hospitalization or death after more than 1 year cannot be excluded.

CONCLUSIONS

Among patients with severe hypoxemia, long-term oxygen therapy used for 24 hours per day, as compared with 15 hours per day, did not reduce the risk of hospitalization or death within 1 year.

LINKS: FULL ARTICLE | NEJM QUICK TAKE | EDITORIAL

FURTHER INFORMATION

Trial registration: ClinicalTrials.gov number, NCT03441204

Trial funding: Crafoord Foundation and others

Full citation: Ekström M, Andersson A, Papadopoulos S, et al. Long-term oxygen therapy for 24 or 15 hours per day in severe hypoxemia. N Engl J Med 2024;391:977-88. DOI: 10.1056/NEJMoa2402638

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